Gagandeep Singh

AI/ML Architect | Generative AI Leader | Data Scientist

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Github HuggingFace Portfolio & Case Studies

PROFILE

Senior AI/ML Engineer with over six years of experience leading the end-to-end design and deployment of production-grade GenAI systems. I specialize in bridging the gap between cutting-edge modeling—including **agentic systems (ReAct, LangChain/LangGraph, MCP), advanced retrieval (RAG, GraphRAG), and model optimization (LoRA, Preference Alignment)**—and the robust **MLOps** (CI/CD, Kubernetes), required to deliver real-world business value. Proven track record building enterprise-used chatbots, call analytics platforms, and NLP pipelines that solve critical business problems and automate processes.

PROFESSIONAL EXPERIENCE

Zykrr Technologies Pvt Ltd, *AVP Data Scientist* May 2024 – Jun 2025 | Gurugram

- Built and launched a multi-agent chatbot using **RAG, ReAct, and VectorDB**, **cutting manual data analysis time by 90%** for enterprise clients by enabling complex natural language queries.
- Designed and deployed a call analytics system using a self-hosted Whisper model, **slashing QA audit time by 83%** (from 3 hours to 30 minutes per call)
- Created a summarization tool that converts thousands of comments into topic-wise insights — **saving an average of 4 hours per report**.
- Automated internal operations with a self-hosted N8N instance, **reducing survey setup time by 87%** (2 hours to 15 mins) and data prep time **by 91%** (1 hour to 5 mins).
- Led LLaMA 3.1 fine-tuning (QLoRA) to create specialized adapters that **outperformed the base model by 30%** on domain tasks and **reduced token costs by 28%**.
- Set up **CI/CD pipelines with HuggingFace hub** to deploy models faster and more reliably, reducing manual deployment errors.

Zykrr Technologies Pvt Ltd, *Senior Manager Data Scientist* May 2023 – May 2024 | Gurugram

- Pioneered the company's **GenAI** strategy by building **3+ high-impact GenAI prototypes**, securing executive buy-in and influencing the roadmap for the flagship **chatbot** product.
- Led initial POC development for predictive analytics features, defining a product direction that contributed to a **15% increase in client upsell opportunities**.
- Collaborated with business teams to translate complex client needs into technical requirements for **predictive analytics**, directly impacting product-market fit.

Zykrr Technologies Pvt Ltd, *Manager Data Scientist* May 2022 – May 2023 | Gurugram, India

- Owned end-to-end development of the **Key Driver Analysis** product, a core feature now used by **70%+ of enterprise clients** and contributing to **$50k in new ARR**.
- Engineered an NLP feature to extract actionable insights from text using **T5** models, **reducing clients' manual analysis time by an average of 10 hours per week**.

- Mentored a junior data scientist and led the team in delivering **over 10 custom enterprise dashboards**.

Zykrr Technologies Pvt Ltd, *Senior Data Scientist* May 2021 – May 2022 | Gurugram

- Containerized ML applications with Docker and orchestrated deployment via Kubernetes, **improving system reliability to 99.95% uptime** and enabling **zero-downtime releases**.
- Reduced inference costs by 50% by replacing **BERT** with **quantized DistilBERT** models with negligible impact on model accuracy.
- Discovered, fixed, and successfully merged a pull request for a critical bug in the **simpletransformers** library, restoring **NER** functionality for the **open-source community**.

Zykrr Technologies Pvt Ltd, *Data Scientist & Intern* Apr 2019 – May 2021 | Gurugram

- Engineered a keyword-level sentiment algorithm that improved accuracy by **30% on complex, multi-topic sentences** and up to **57% on simpler text.**
- Migrated the legacy analytics engine from TF-IDF to BERT, boosting the core model's F1-score by 18% and **increasing robustness against unseen data through Adversarial and Behavioral testing**.
- Built the foundational data pipeline using Redis and AWS, creating a robust queue capable of processing **over 10,000 text records per hour** with zero data loss.

PERSONAL PROJECTS

InnerVoice Jun 2025 – present

An AI assistant that watches your screen and builds a knowledge graph around what you're doing. Think of it as a quiet observer that understands your tasks and offers help when it makes sense.

CHAOS Framework for Agentic AI Jun 2025 – Jun 2025

A framework to generate synthetic data for training AI agents to think more like humans — with reasoning, confidence, and adaptability. It's an ongoing experiment in how far we can push LLMs beyond simple Q&A.

Talk with Figma using Claude Desktop May 2025 – Jun 2025

Enables Claude Desktop App to control Figma through the MCP (Model Context Protocol) and WebSocket communication. This project was **recognized and featured in the official Model Context Protocol (MCP) GitHub repository**.

EDUCATION

GL Bajaj Institute of Technology and Management, *BTech* Aug 2015 – Jun 2019 | Greater Noida

SKILLS

— **Programming Languages**: Python | **Frameworks & Libraries**: PyTorch, HuggingFace, FastAPI, MLflow, Opik, Langgraph, Langchain, CrewAI, AutoGen, N8N. | **Machine Learning**: Natural Language Processing, Generative AI, Predictive Analytics, Speech Analytics | **Cloud & DevOps**: Docker, Kubernetes, Azure, AWS | **Databases**: SQL, Vector Databases (Qdrant, Milvus, Pinecone, ChromaDB) | **Additional Skills**: AI Eval, Prompt Management